AI on the horizon

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1. Reinforcement Learning

RL for sequential (clinical) decision-making

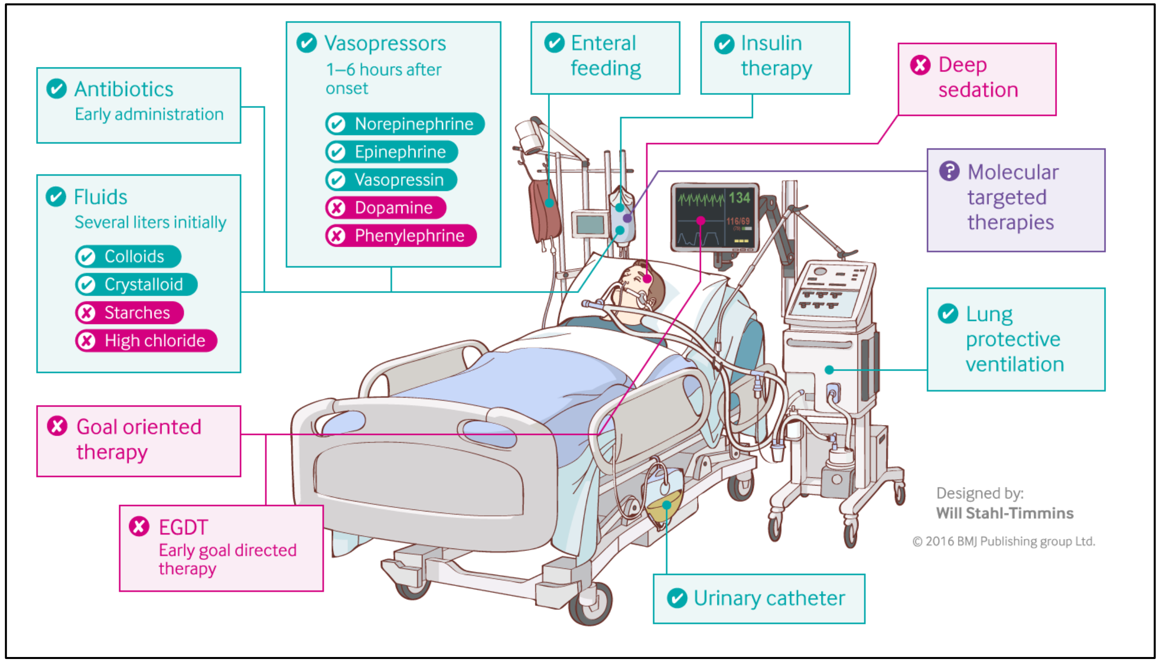
An agent

takes actions

based on the state of a system

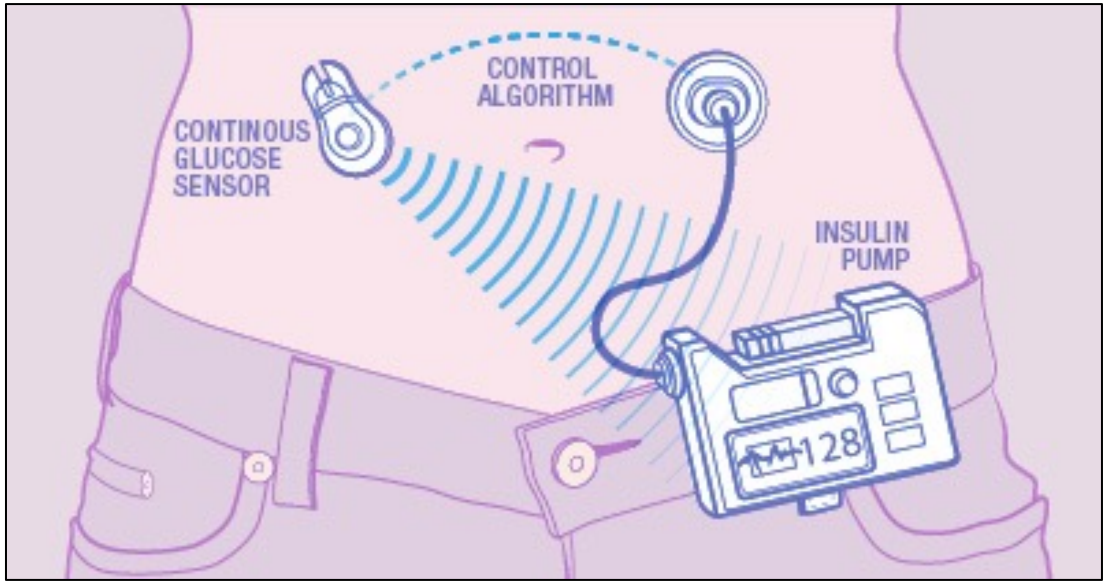
to maximize reward

RL for sepsis management



An agent	A clinician
takes actions	gives fluid and/or vasopressor
based on the state of a system	based on the patient's physiologic status
to maximize reward	to maximize chance of survival

RL for close—loop BG control



An agent	A computer program
takes actions	administers insulin
based on the state of a system	based on blood glucose and recent patient behaviors
to maximize reward	to maintain normoglycemia

In RL, we typically learn “from scratch”:

- Try things and see what works
- Initially our actions are random

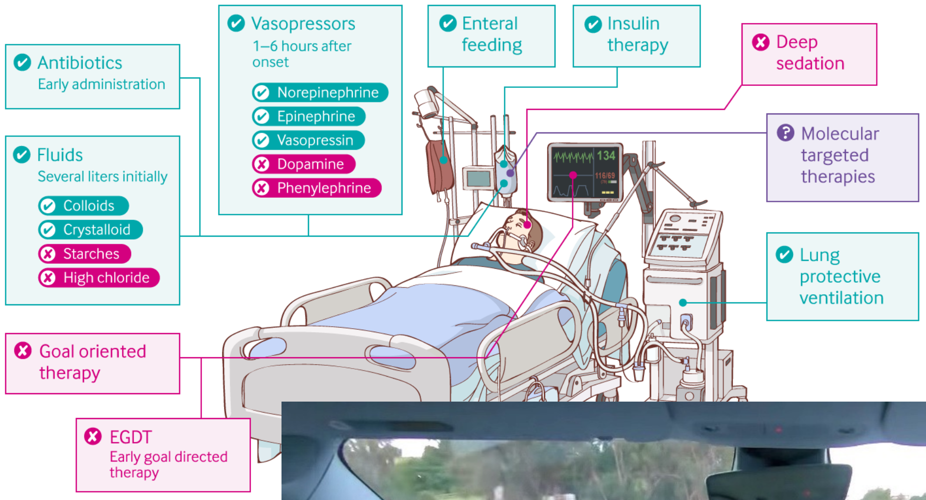
Drone Uses AI and 11,500 Crashes to Learn How to Fly

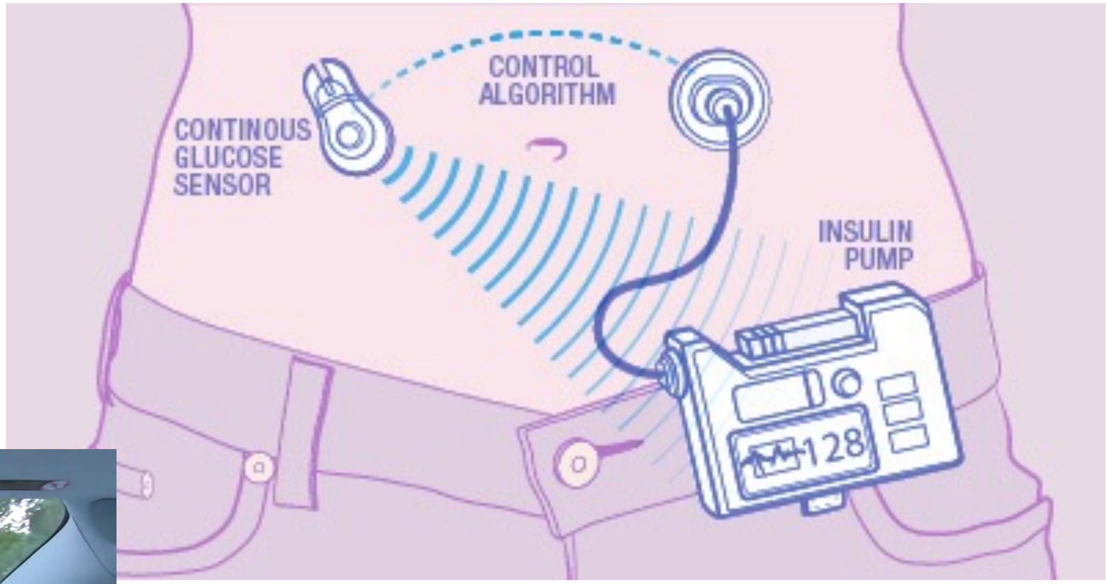
Crashing into objects has taught this drone to fly autonomously, by learning what not to do

By [Evan Ackerman]

Failing 11,500 times isn't always an option

Treating sepsis: the latest evidence





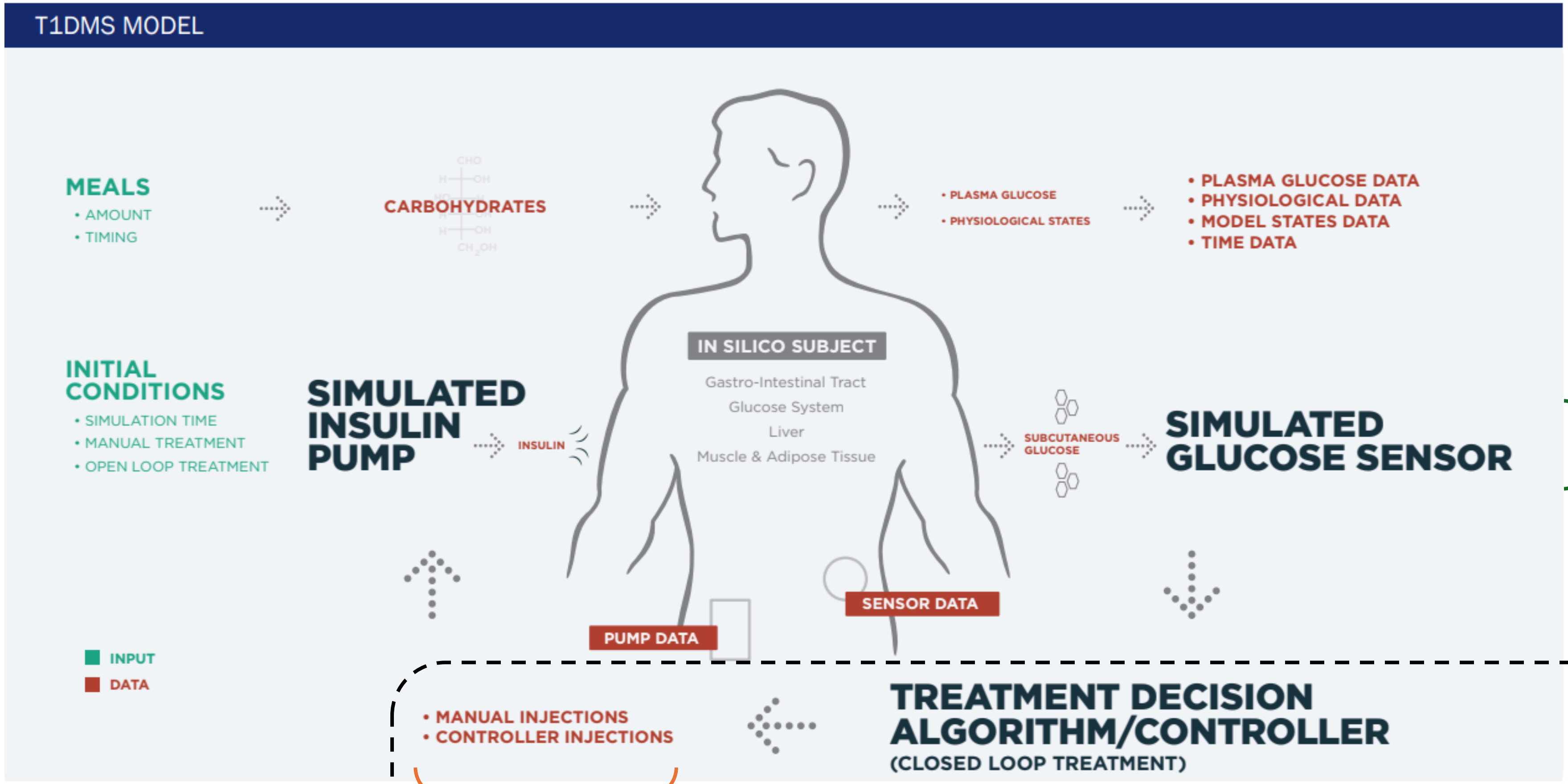
-> What are the alternatives?

A1. Learn from Observational Data

The eICU Collaborative Research Database, a freely available multi-center database for critical care research. Pollard TJ, Johnson AEW, Raffa JD, Celi LA, Mark RG and Badawi O. Scientific Data (2018).

- provides a large number of (state, action, reward) examples

A2. Learn Policy from Simulated Environment



state

actions

Reinforcement Learning Agent

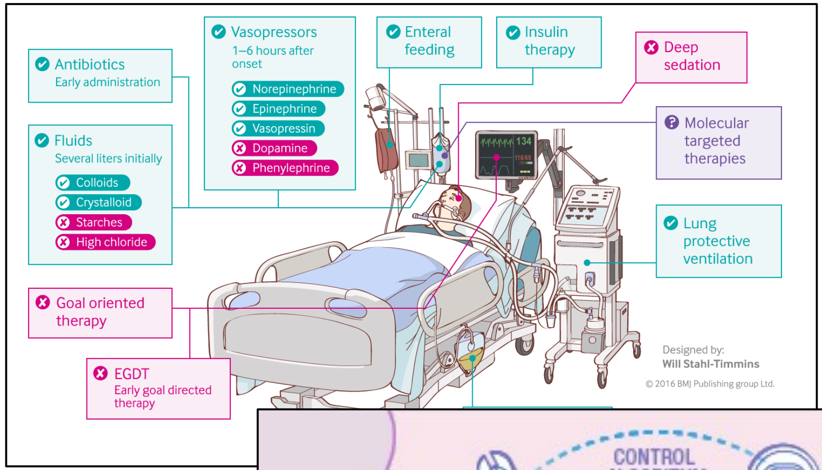
A3. Learn with Expert Oversight

physiologic state

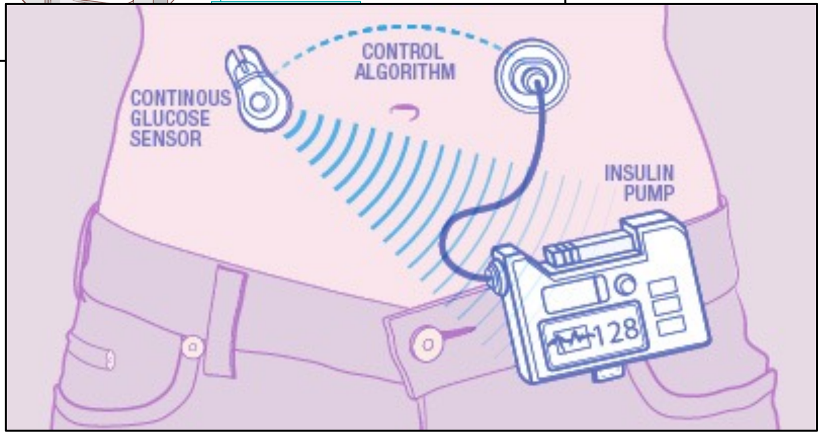
algorithm recommendation

physician approval

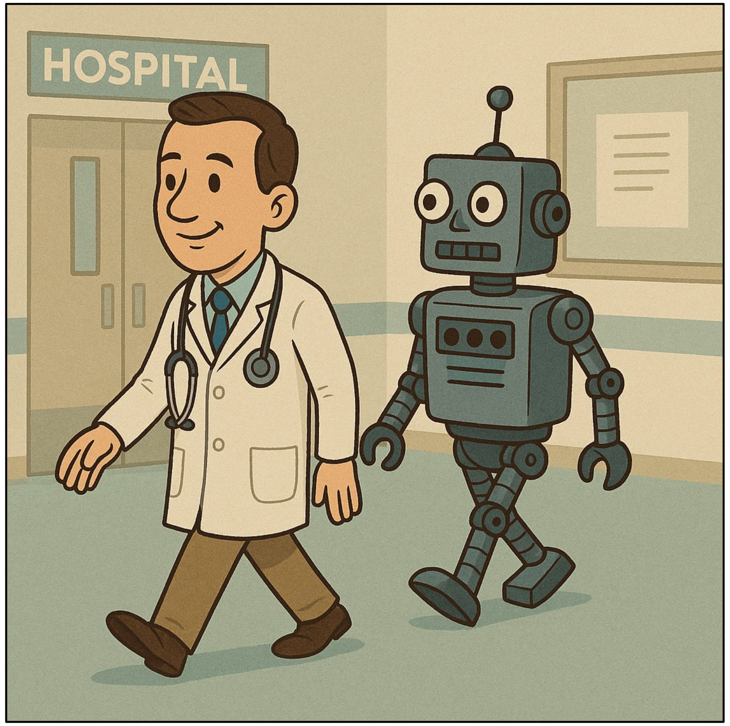
Progress in safe RL and imitation learning



sepsis management



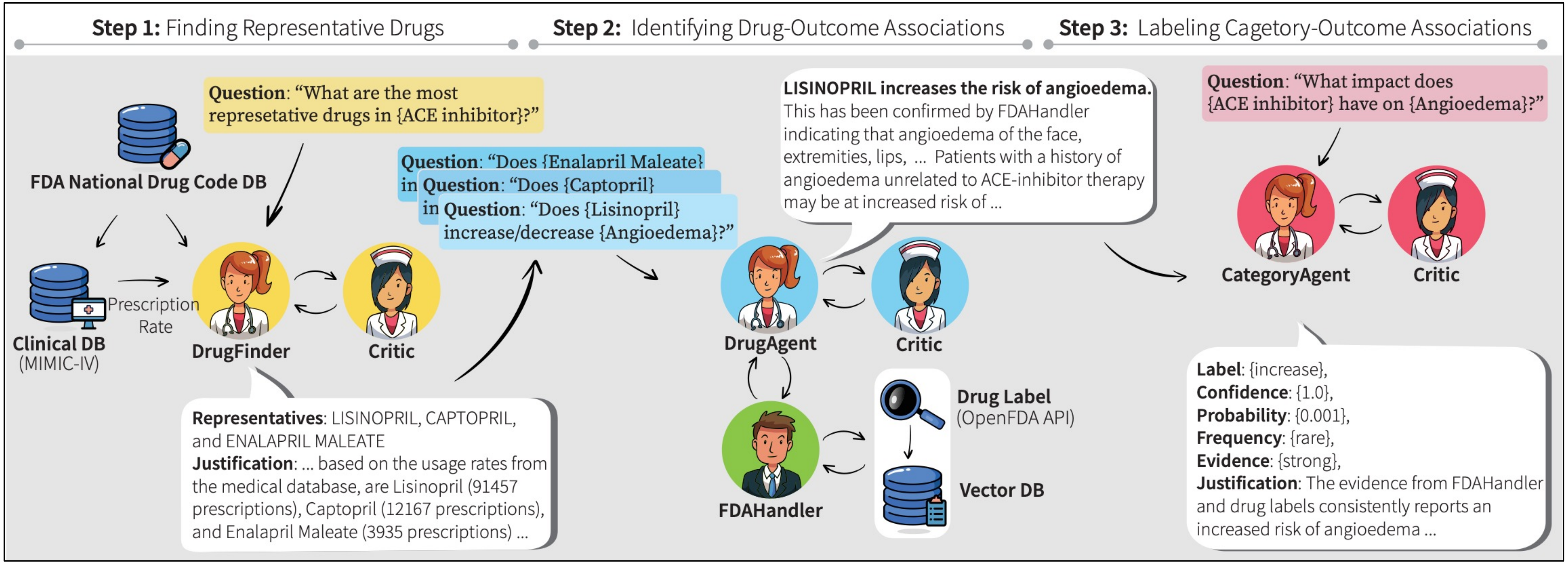
closed-loop BG control



safe learning by imitation

2. Agentic AI

Agents, tools, and orchestration



Jihye Choi, Nils Palumbo, Prasad Chalasani, Matthew Engelhard, Somesh Jha, Anivarya Kumar, & David Page, (2024). *MALADE: Orchestration of LLM-powered Agents with Retrieval Augmented Generation for Pharmacovigilance.* Machine Learning for Healthcare 2024.

Automated CEU Eligibility Assessment

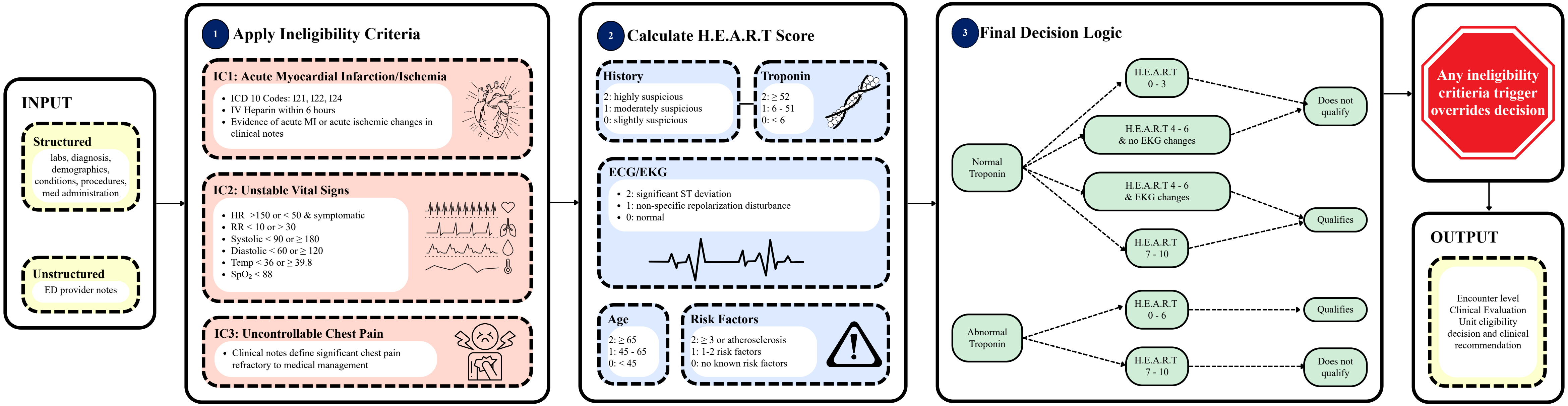


Figure 1: Rapid Automated Assessment of CEU Eligibility (RAACE) pipeline overview.

A few more discussion questions

How do we ensure clinical AI remains human-centered?

- Should we insist that human providers make the decisions?
- How do we ensure that care remains aligned with our values?

What influence do AI companies have on healthcare?

- Should we be pushing back against this?
- If so, how?